Signal Estimate

(HANDOUTS)

Signal Officer Advance Course (SOAC) Subject 8.2 • Signal Officer Basic Course (SOBC) Subject 8.1

Courses / Subjects	SOAC – Module 8 (Signal Planning), Subject 8.2 Signal Estimate SOBC – Module 8 (Signal Plans and Orders), Subject 8.1 Signal Estimate Development
References	PAM 6-0102 Signal Operations (15 April 2022): paras 0216, 0302–0305, 0314 PAM 6-00 Command and Control Communications, and Cyber Systems (2nd Edition, 11 December 2017): Chapters II and III PAM 3-00 Army Operations (06 May 2014): para 0524 (METT-TC) and para 0609 (running estimates) Supporting: PAM 3-0201 Commander and Staff Organization and Operations (29 December 2023): Chapter III (problem solving), Chapter VII (running estimates), Chapter IX (MDMP)
How to use this handout	Read Parts I to III before the session. Parts IV and V are covered in class. Part VI is the practical exercise (Exercise BANTAY LINYA, at the COA analysis stage), and Part VII holds the worksheets. SOAC groups produce a complete signal estimate; SOBC groups produce the basic inputs to one.

Introduction

Before the signal officer can plan, the signal officer must estimate. PAM 6-0102 says the signal estimate **"sets the stage for signal support operations"**: it lets the signal officer understand the environment the unit will operate in, its effect on signal equipment, and the signal support strategy to apply (para 0216). The quality of the signal plan "relies heavily on the quality of signal estimates conducted before the actual planning" (para 0217). This handout explains what the signal estimate is, how it is made, the sequence in which it is written, and how it helps the commander compare courses of action.

Learning Objectives

Signal Officer Advance Course – Subject 8.2

TLO. At the end of this session, the students will be able to create a Signal Estimate for the effective employment of all available signal communications during signal operations IAW PAM 6-0102-03. (C5)

ELO	Enabling Learning Objective	Covered in
8.2.1	Discuss Signal Estimate (C3)	Parts I–II
8.2.2	Discuss the steps in making a Signal Estimate (C3)	Part III
8.2.3	Outline the sequence of Signal Estimate (C4)	Parts IV–V

Signal Officer Basic Course – Subject 8.1

TLO. At the end of the session, the students will be able to prepare basic inputs to a signal estimate to support operational planning and decision-making in accordance with PAM 6-00 and PA Signal Doctrine. (P3)

ELO	Enabling Learning Objective	Covered in
8.1.1	Describe METT-TC in signal planning (C3)	Part II
8.1.2	Identify signal requirements based on mission variables (C2)	Part II

ELO	Enabling Learning Objective	Covered in
8.1.3	Prepare basic components of a signal estimate (P3)	Parts III–IV
8.1.4	Assist in comparing courses of action (COAs) based on signal support considerations (P3)	Part V

Definition of Terms

Term	Meaning
Signal estimate (C4S estimate)	An orderly and logical examination of all factors, known or assumed, that may affect the successful accomplishment of the signal support operations mission (PAM 6-0102, paras 0302–0303).
Running estimate	A staff section's continuous assessment of current and future operations to determine whether the current operation is proceeding as intended and whether future operations are supportable (PAM 3-00, para 0609). The signal estimate is the S6's running estimate.
METAL	Memory aid for the signal estimate: Mission, Enemy signal capabilities, Troops available, Area of operations, Logistical requirements (PAM 6-0102, para 0303).
METT-TC	The mission variables: Mission, Enemy, Terrain and weather, Troops and support available, Time available, Civil considerations (PAM 3-00, para 0524).
Fact / assumption	A fact can be verified. An assumption is believed true but cannot yet be verified; it must be valid (likely true) and necessary (needed to continue) (PAM 3-0201, paras 0309–0310).
Course of action (COA)	A broad potential solution to an identified problem (PAM 3-0201, para 0981). In the signal estimate, the S6 analyzes the unit's COAs for signal supportability and the signal means or concepts that would support each.
Screening / evaluation criteria	Screening criteria test whether a solution is acceptable at all; evaluation criteria distinguish one solution from another (PAM 3-0201, paras 0320–0322).
C4S	Command and control communications, and cyber systems.

Part I. The Signal Estimate

1. Definition and scope

The signal estimate is **an orderly and logical examination of all factors, known or assumed, that may affect the successful accomplishment of the signal support operations mission** (PAM 6-0102, para 0302). It may be formed in two ways:

- **Separate estimates** for each signal support service and C4S capability, which the signal officer then integrates; or
- **A single estimate** that considers all means required for signal support operations.

Either way, it is **revised as the situation changes and new facts are introduced** (para 0302). PAM 6-0102 also calls it the **C4S estimate** (para 0303).

2. Why it matters

- It **sets the stage** for signal support operations and determines the signal support strategy (PAM 6-0102, para 0216).
- The **signal plan is only as good as the estimate** done before it (para 0217).
- It is **dynamic**: like combat operations, it changes from time to time (para 0216).
- It is how the signal officer **advises the commander and the S3**: the signal planner recommends items for the unit's plans and orders in the form of a signal estimate (para 0314).

3. The signal estimate as a running estimate

Every staff section keeps a running estimate (PAM 3-00, para 0609; PAM 3-0201, Chapter VII). The signal estimate is the S6's. It is started on receipt of the mission, refined during mission analysis, used to develop and test COAs in the war game, used to compare COAs, and becomes the basis of Annex H. Significant details in the running estimate may become enclosures to the order in the functional annexes (PAM 3-0201, para 0710).

MDMP step	What the signal estimate provides
1. Receipt of mission	Current C4S status: equipment, personnel, nets in use
2. Mission analysis	Signal tasks, facts, assumptions, shortfalls, EEFIs for the mission analysis brief
3. COA development	Signal means or concept that could support each COA
4. COA analysis	C4S needs compared with available assets for each COA (PAM 3-0201, para 09155)
5. COA comparison	Ranking of COAs for signal supportability; recommendation
6–7. Approval and orders	Basis of Annex H and paragraph 5

Part II. METT-TC, METAL and Signal Requirements

1. METT-TC in signal planning

PAM 3-00 describes the mission variables, METT-TC, as the major variables considered during mission analysis (para 0524). The signal officer applies each one to communications:

Variable	What it means (PAM 3-00)	Signal questions
Mission	The task and purpose; specified and implied tasks; intent two levels up	What signal tasks are specified or implied? Which nets must we enter or provide?
Enemy	Dispositions, doctrine, equipment, capabilities, vulnerabilities, probable COAs	Can the enemy intercept, jam, direction-find, attack sites, use cyber or drones?
Terrain and weather	OAKOC; climate and weather effects	Line of sight, dead spaces, foliage, propagation, storms, site access?
Troops and support	Number, type, capability and condition of friendly troops and support	Equipment and personnel status; support from higher and adjacent units?
Time available	Time to plan, prepare and execute; time for subordinates	Time to move, install, test and rehearse?
Civil considerations	ASCOPE: areas, structures, capabilities, organizations, people, events	Commercial networks, power, NTC rules, PNP and LGU, crowded events?

2. METAL – the memory aid for the signal estimate

PAM 6-0102 gives the keyword **METAL** to help the signal planner prepare the estimate "in any given situation and location" (para 0303):

Letter	Factor	Consider	Related METT-TC
M	Mission	The mission in precise terms; specified and implied signal tasks	Mission; Time
E	Enemy	Enemy signal capabilities: EW, interception, cyber, attacks on C4S sites	Enemy
T	Troops	Signal troops and equipment available to accomplish the mission	Troops and support
A	Area of operations	Terrain, weather, propagation, existing communications, civil factors	Terrain and weather; Civil considerations
L	Logistical	Supply, maintenance, power, batteries, transport, funds	Troops and support

> **Note on the memory aid**
> Some older handouts write the keyword as "MENTAL." The current reference, PAM 6-0102 para 0303, uses **METAL**.

3. Identifying signal requirements from the mission variables

Requirements come from five sources (PAM 6-0102, para 0305): the **commander's guidance or intent**; **staff guidance** (the staff are the users of signal support); **higher, subordinate and adjacent units**; **SOPs**; and the **CEOI and CESI**. Use the METT-TC questions above to turn each source into specific requirements: who needs to talk to whom, by what means, where and when.

Part III. Steps in Making the Signal Estimate

An estimate is a problem-solving process. PAM 3-0201 describes the Army's problem-solving steps: amass information and knowledge, identify the issue, create criteria, generate possible solutions, analyze them, compare them, and make and implement the decision (para 0307). For the signal estimate, these are applied in five steps:

Step	What the signal officer does	Problem-solving step (PAM 3-0201)
1. Know the problem	Analyze the unit's mission and the commander's intent; determine specified and implied signal tasks; state the signal mission	Identify the issue
2. Gather data and determine possible solutions	Assemble the facts and assumptions (METT-TC, METAL); determine the signal means or concepts that could support each unit COA; screen them	Amass information; create criteria; generate possible solutions
3. Analyze the solutions	Determine the advantages and disadvantages of each solution against the METAL factors	Analyze possible solutions
4. Compare the solutions	Compare against evaluation criteria; rank in order of preference	Compare possible solutions
5. Recommend or decide	Translate the best solution into a recommendation to the commander (a decision if the commander makes the estimate)	Make and implement the decision

Facts and assumptions (PAM 3-0201, paras 0309–0311)

Facts are verifiable. Assumptions are needed when facts are missing; a good assumption is **valid** (likely true) and **necessary** (needed to continue). Replace assumptions with facts as information comes in, and drop any assumption that does not affect the solution.

Screening criteria (PAM 3-0201, para 0321)

Criterion	Question for a signal solution
Suitable	Does it solve the signal problem, and is it legal (NTC, COMSEC policy) and ethical?
Feasible	Can it be done with the radios, operators, power and funds available?
Acceptable	Are the costs and risks worth it?
Distinguishable	Is it significantly different from the other solutions?
Complete	Does it cover all critical requirements from start to finish?

Part IV. Sequence of the Signal Estimate

The signal estimate is written in a logical sequence. PAM 3-0201 gives the base running estimate format (Figure 7.1), which each staff element tailors to its functional area. Applied to C4S, the sequence is:

Paragraph	Content for the signal estimate
1. Situation and considerations	a. Area of interest – factors of the area of interest that affect C4S b. Characteristics of the area of operations: terrain and weather (propagation, line of sight, critical periods); enemy (EW units, congested parts of the spectrum, C4S capabilities, weaknesses); friendly C4S resources (committed and uncommitted signal elements, equipment status, training and morale, vulnerability to EW); civil considerations (existing commercial systems and the security of using them) c. Facts and assumptions
2. Mission	The signal mission, deduced from the unit's restated mission: what signal support will be provided, including any unusual or special support
3. Courses of action	a. Friendly COAs being war-gamed, with the signal means or concept for each b. Enemy actions that affect C4S c. Evaluation criteria (all staff use the same criteria)
4. Analysis	Each COA analyzed against the criteria and the METAL factors; issues, risks and deficiencies for C4S
5. Comparison	COAs compared and ranked for each key C4S consideration, using a decision matrix
6. Recommendations and conclusions	a. The most supportable COA from the C4S view b. Prioritized issues, deficiencies and risks, with recommendations to mitigate them

Adapted from PAM 3-0201, Figure 7.1 (Generic Base Running Estimate Format), with signal content from PAM 6-0102

Relation to the traditional five-paragraph C-E estimate

Traditional C-E estimate	Running estimate (PAM 3-0201)
1. Mission	2. Mission
2. Situation and courses of action	1. Situation and considerations; 3. Courses of action
3. Analysis of courses of action	4. Analysis
4. Comparison of courses of action	5. Comparison
5. Recommendation / decision	6. Recommendations and conclusions

Analysis questions by METAL

Factor	Questions for each COA
Mission	How well does it accomplish the signal mission? Is support timely?
Enemy	Which critical points can the enemy affect (EW, cyber, site attack)? What effect does the COA have on the enemy's capabilities?
Troops	Does it use signal units effectively? Do they meet the requirements? Is there time to move and install?
Area	Does it make the best use of terrain, line of sight and existing systems? Are key terrain features for signal sites considered?
Logistical	Are supplies, batteries, power, maintenance and transport adequate? Can it be done with organic equipment, or is special equipment needed?

Comparison factors

Usually an advantage	Usually a disadvantage
• Economy of personnel and facilities – leaves uncommitted assets for continuous operation • Signal sites that meet tactical needs and reduce exposure to enemy ground, air and EW attack • Redundant, independent means (PACE)	• Committing all personnel and facilities early in the operation • A system highly vulnerable to EW during critical phases • Failure to meet the tactical requirement – reject such a COA

The recommendation

State which COA is most supportable and what the signal elements of the command will do to support it: **who, what, when, where, how and why**. When a staff officer prepares the estimate, the last paragraph is a recommendation; when the commander makes it, it is a decision.

Part V. Comparing COAs on Signal Support Considerations

In Step 5 of the MDMP, each staff officer compares the COAs from his or her functional area, often with a decision matrix built on the evaluation criteria set before the war game (PAM 3-0201, paras 09172–09177). The S6's comparison goes into paragraphs 5 and 6 of the signal estimate. Ranks are 1 = best; multiply by the weight and add; **the lowest total is preferred**.

C4S criteria →	Continuity of Brigade C2	Coverage of patrols	Demand on signal resources	Vulnerability to EW and site attack	Total
Weight	2	2	1	1	
COA A	1 (2)	2 (4)	2 (2)	1 (1)	6 (9)
COA B	2 (4)	1 (2)	1 (1)	2 (2)	6 (9)

Illustration only: when totals tie, the S6 compares criterion by criterion and explains the trade-off to the commander. The matrix supports the recommendation; it does not make it.

Self-Check Questions

1. Define the signal estimate according to PAM 6-0102.
2. What are the two ways a signal estimate may be formed?
3. Why does PAM 6-0102 say the signal estimate "sets the stage" for signal support operations?
4. What does METAL stand for?
5. Give one signal question for each METT-TC variable.
6. What are the five sources of signal requirements?
7. List the five steps in making the signal estimate.
8. What makes an assumption valid and necessary?
9. List the six paragraphs of the signal estimate (running estimate format).
10. Name two factors that are usually advantages and two that are usually disadvantages when comparing signal COAs.
11. In a decision matrix, is the lowest or highest total preferred?
1. When is the last paragraph of an estimate a recommendation, and when is it a decision?

Summary

The signal estimate is an orderly and logical examination of all factors that may affect the signal mission. It sets the stage for signal support, is kept as the S6's running estimate, and is revised as the situation changes. The signal officer makes it in five steps – know the problem, gather data and determine possible solutions, analyze, compare, recommend – using METT-TC and the METAL memory aid. It is written in a logical sequence: situation and considerations, mission, courses of action, analysis, comparison, and recommendations and conclusions. Its comparison of COAs on signal support considerations helps the commander choose a plan that can actually be commanded and controlled.

References

- PAM 6-0102 Signal Operations, promulgated 15 April 2022
- PAM 6-00 Command and Control Communications, and Cyber Systems, 2nd Edition, 11 December 2017
- PAM 3-00 Army Operations, 06 May 2014
- PAM 3-0201 Commander and Staff Organization and Operations, 29 December 2023 (supporting)
- Program of Instruction, Signal Officer Advance Course, Subject 8.2, and Signal Officer Basic Course, Subject 8.1